CSP-203: Software Tools & Technologies Lab



Plotting Tools: GNU Plot
(Lab Session – 15)

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Multiplot Example:

multiplot.gp
~/

```
# Set terminal and output
set terminal pdfcairo size 20cm, 16cm enhanced font "Times New Roman,12"
set output "multiplot.pdf"

# Enter multiplot mode
set multiplot title "Manual Subplots Layout"

### First subplot: Top full-width
set origin 0, 0.5            # Bottom-left corner at (0, 0.5)
set size 1, 0.5              # Full width, half height
set title "Top Plot"
set xlabel "x"
set ylabel "sin(x)"
plot sin(x) with lines title "sin(x)"

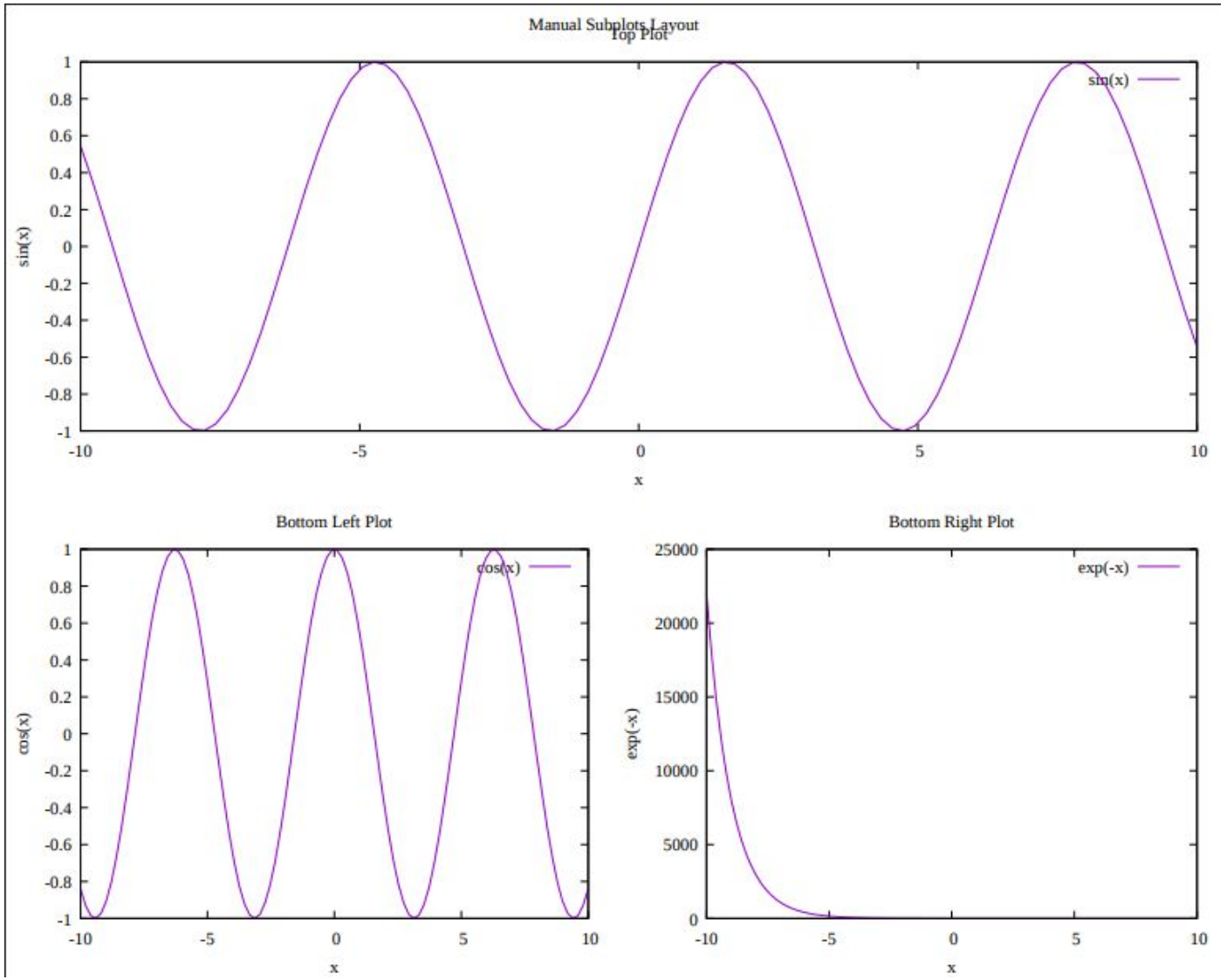
### Second subplot: Bottom-left
set origin 0, 0              # Bottom-left corner at (0, 0)
set size 0.5, 0.5            # Half width, half height
set title "Bottom Left Plot"
set xlabel "x"
set ylabel "cos(x)"
plot cos(x) with lines title "cos(x)"

### Third subplot: Bottom-right
set origin 0.5, 0            # Bottom-left corner at (0.5, 0)
set size 0.5, 0.5            # Half width, half height
set title "Bottom Right Plot"
set xlabel "x"
set ylabel "exp(-x)"
plot exp(-x) with lines title "exp(-x)"

# Exit multiplot mode
unset multiplot
unset output

```

Output:


Manual Subplots Layout
Top Plot
sin(x)
x
Bottom Left Plot
cos(x)
x
Bottom Right Plot
exp(-x)
x

❑ 2×2 Grid

```
set multiplot layout 2,2 title "2x2 Multiplot Example"

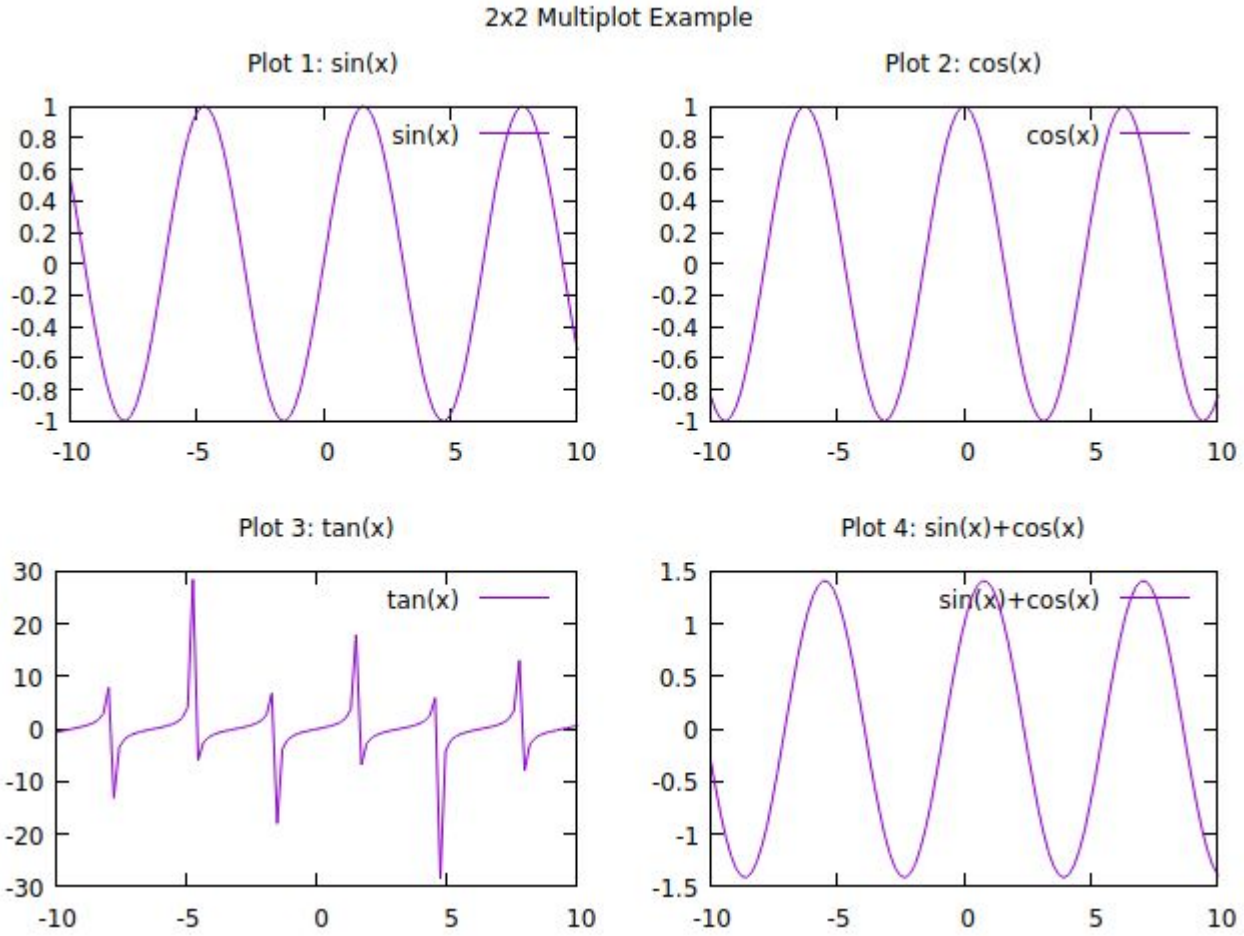
set title "Plot 1: sin(x)"
plot sin(x)

set title "Plot 2: cos(x)"
plot cos(x)

set title "Plot 3: tan(x)"
plot tan(x)

set title "Plot 4: sin(x)+cos(x)"
plot sin(x)+cos(x)

unset multiplot
pause -1 "Press Enter to close"
```



❑ Bars / Histogram

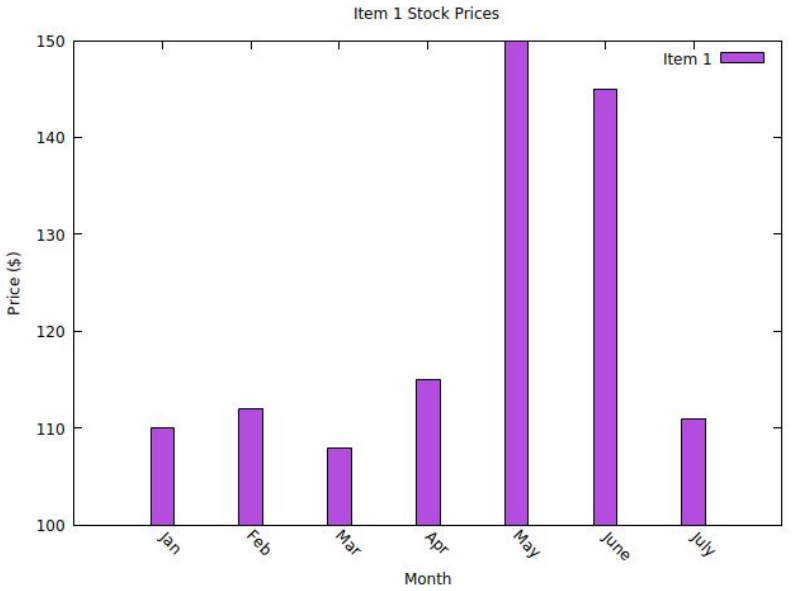
```
set style data histograms
set style fill solid 0.7 border -1   # solid bars, 70% opaque
set boxwidth 0.8 relative            # control bar width
```

If you make it `1.0`, bars touch each other.

```
set title "Item 1 Stock Prices"
set xlabel "Month"
set ylabel "Price ($)"
set xtics rotate by -45
set yrange [100:145]

plot "file2.dat" using 2:xtic(1) title "Item 1"

pause -1
```



Jan 110
Feb 112
Mar 108
Apr 115
May 50
June 45
July 11

file2.dat

Plotting a barplot:

- set style data histograms : Instructs GNUplot to treat the data as histogram, rather than a line or scatter plot.
- set style fill pattern 4 border -1 : Pattern fills the bar with predefined pattern, border -1 disables the black border for each bar.
- set boxwidth 0.6 : Sets the relative width of the bars.
- plot 'item1.dat' using 2:xtic(1) title "Item 1 Stock prices" lc rgb "red"

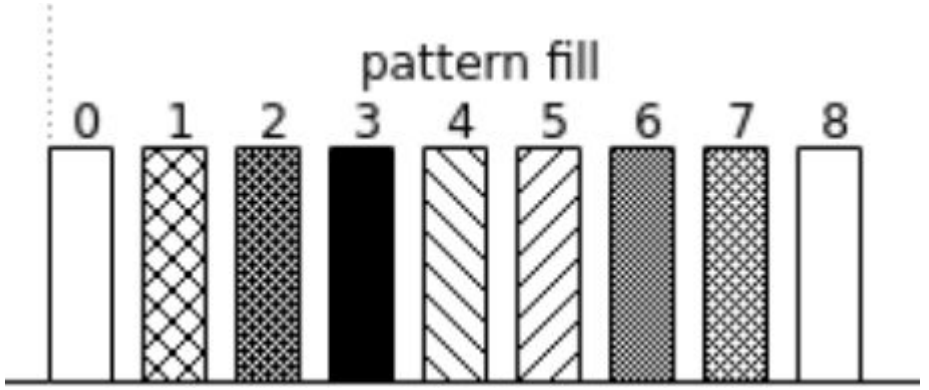
  [2:xtic(1), Column 2 is used for the height of the bars (the expenses). Column 1 is used for the X-axis labels (the month names)].

Different border setting:

Command	Meaning
border -1	No border at all (completely removes the outline)
border 0	Draws only the **outline (border)** of the bars, but no top/bottom lines
border 1 *(default)*	Draws a full border: all 4 sides of the bar

❑ Style for Bar:

```
set style fill empty                        # hollow boxes
set style fill solid 1.0 border             # solid fill, with border (default color)
set style fill solid 0.6 noborder           # semi-dense fill, no border
set style fill transparent solid 0.5        # 50% transparent fill (if terminal supports it)
set style fill pattern 2 border 1 lc rgb "black"  # pattern fill with black border
```



Example Code:

```
set terminal pdfcairo size 20cm,14cm font "Times New Roman,12"
set output "monthly_expenses.pdf"
set title "Monthly Expenses"

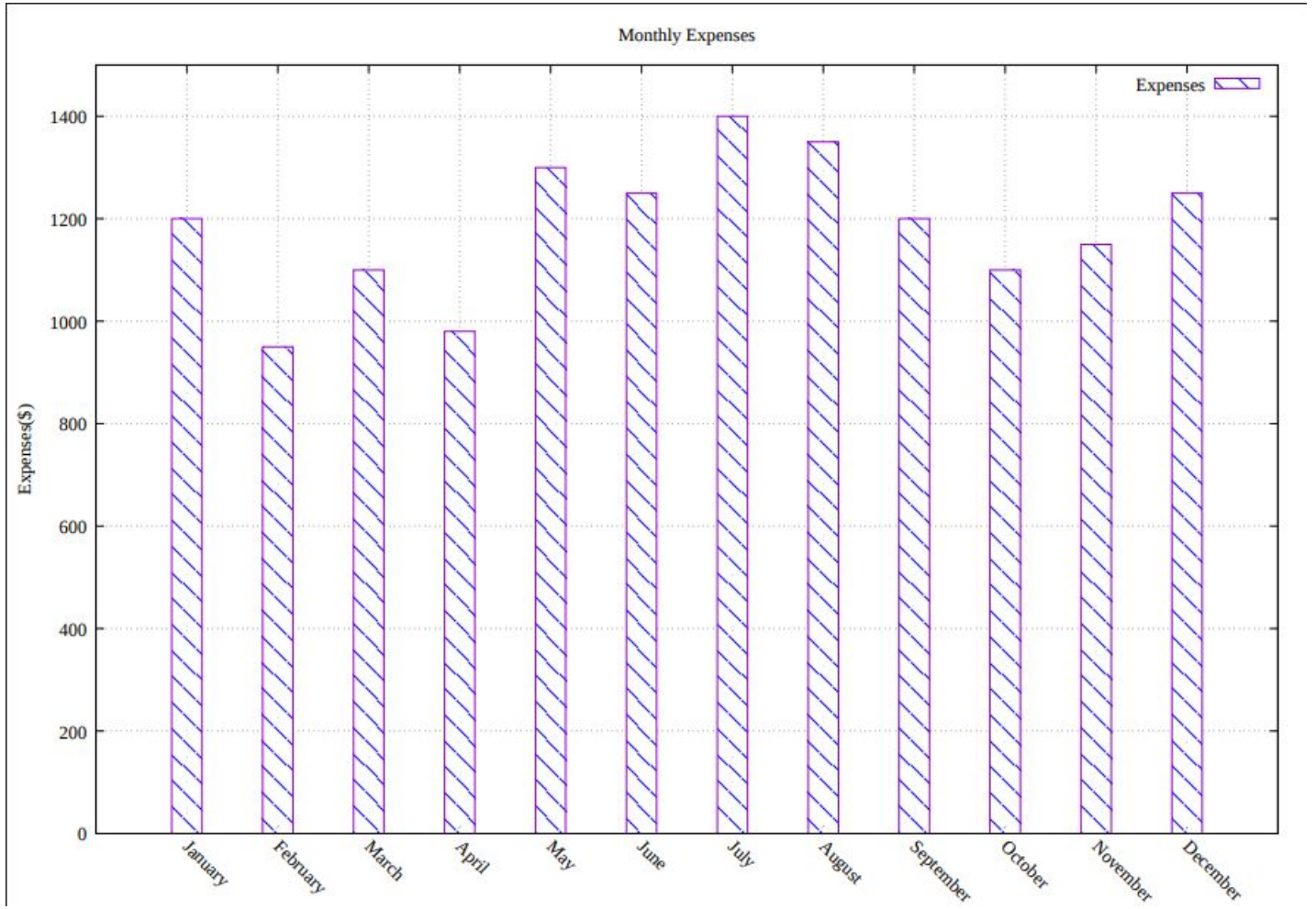
set style data histogram
set style fill pattern 4 border 1
set boxwidth 0.8

set xtics rotate by -45
set xlabel "Month"
set ylabel "Expenses ($)"
set yrange [100:170]

set grid
plot "expenses.dat" using 2:xtic(1) title "Expenses" lc rgb "blue"

unset output
unset terminal
```

Output:

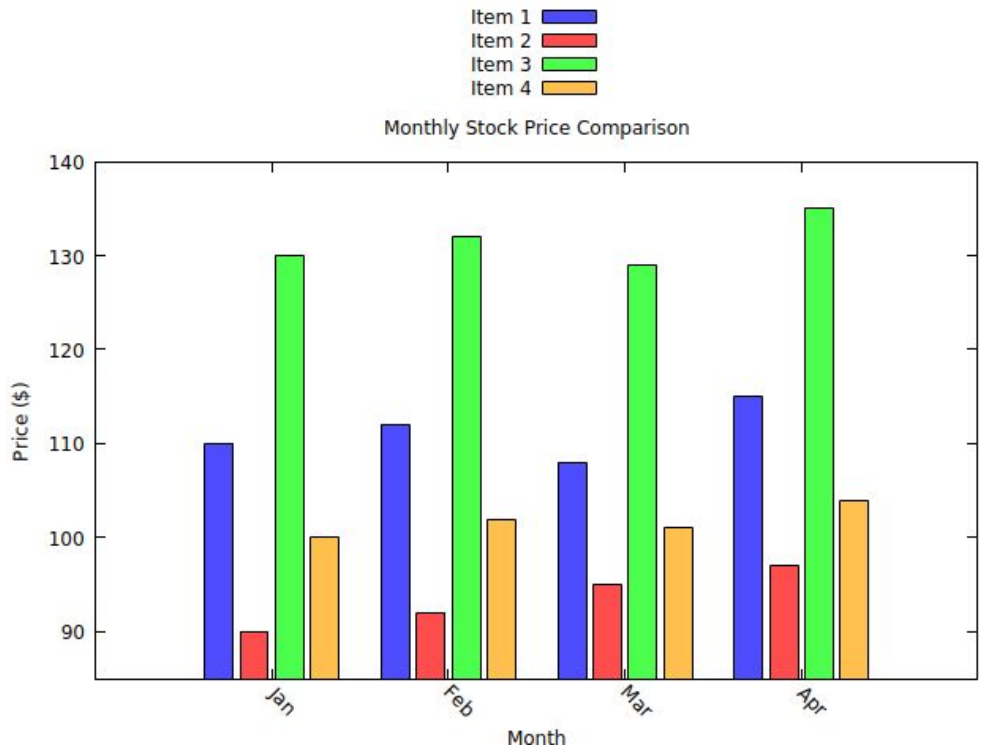

Monthly Expenses
Expenses
Expenses($)
0
200
400
600
800
1000
1200
1400
January
February
March
April
May
June
July
August
September
October
November
December

❑ Multibar

```
set style data histograms
set style histogram cluster gap 1
set style fill solid 0.7 border -1
set boxwidth 0.9 relative

set title "Monthly Stock Price Comparison"
set xlabel "Month"
set ylabel "Price ($)"
set key outside top center
set xtics rotate by -45

plot "stock.dat" using 2:xtic(1) title "Item 1" lc rgb "blue", \
    "" using 3 title "Item 2" lc rgb "red", \
    "" using 4 title "Item 3" lc rgb "green", \
    "" using 5 title "Item 4" lc rgb "orange"
```



Month	Item1	Item2	Item3	Item4
Jan	110	90	130	100
Feb	112	92	132	102
Mar	108	95	129	101
Apr	115	97	135	104

Quick reference cheat-sheet

- `set style data histograms` — turn on histogram (bar) mode. gnuplot.sourceforge.net
- `set style histogram clustered | rowstacked | columnstacked | errorbars` — choose layout. Massachusetts Institute of Technology
- `set style fill {empty|solid <density>|transparent solid <alpha>|pattern <n>} {border|noborder}` — fill options (use `noborder` to remove). gnuplot.info
- `set boxwidth <w> {absolute|relative}` — width in data units or fraction of slot; default = auto (touching). gnuplot.sourceforge.net

❑ HeatMap

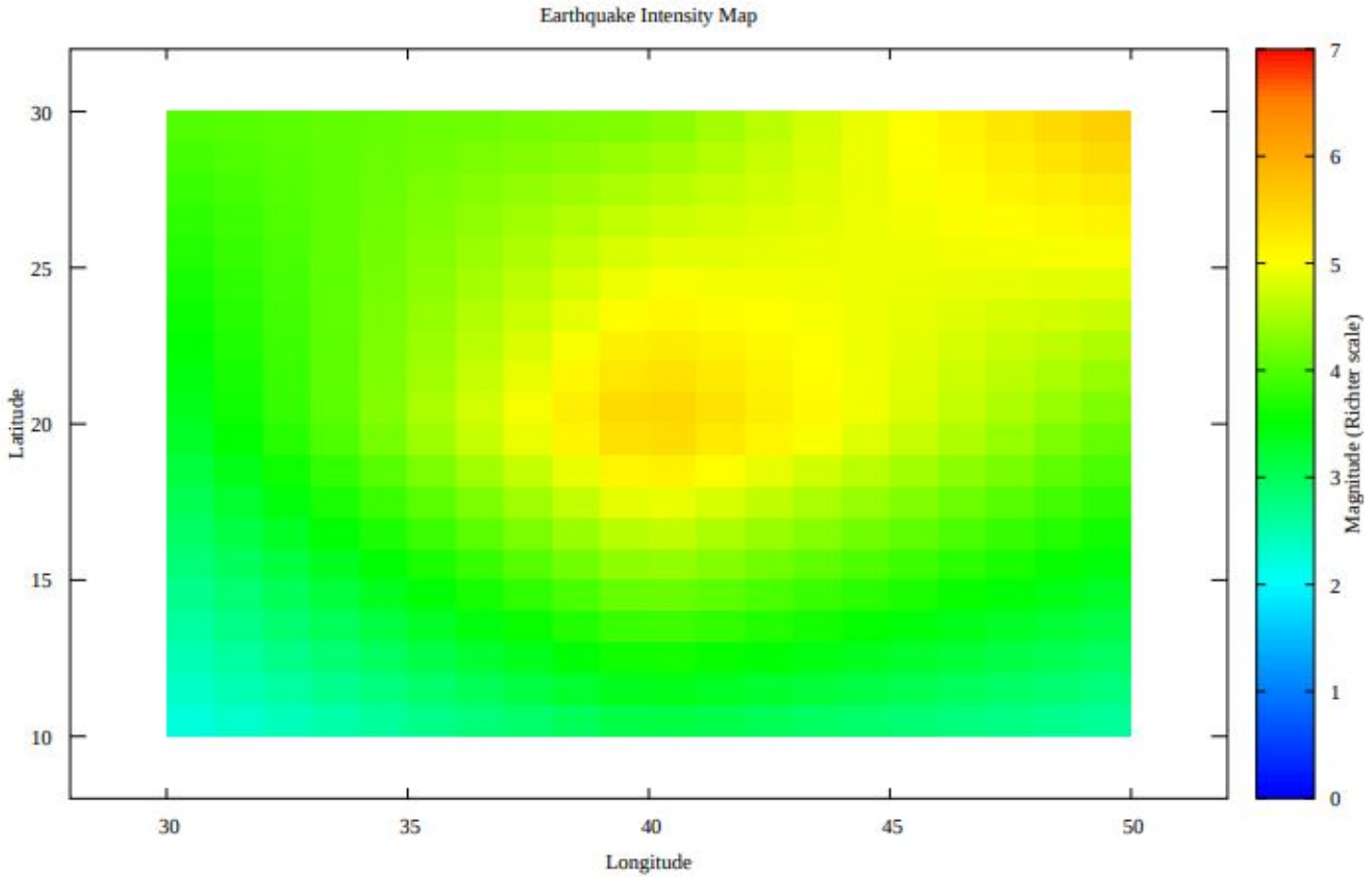
▯ Data Given:

Longitude Latitude Earthquake_intensity
30.0 10.0 2.1
30.0 20.0 3.2
30.0 30.0 4.0

40.0 10.0 3.0
40.0 20.0 5.6
40.0 30.0 4.2

50.0 10.0 2.5
50.0 20.0 4.1
50.0 30.0 5.7

Output:


Earthquake Intensity Map
Latitude
Longitude
Magnitude (Richter scale)
30
25
20
15
10
30
35
40
45
50
7
6
5
4
3
2
1
0

- Sample Code of Heatmap:

```
set terminal pdfcairo size 20cm,14cm enhanced font 'Times New Roman,12'
set output 'quake_heatmap.pdf'
set title "Earthquake Intensity Map"
set xlabel "Longitude"
set ylabel "Latitude"

set cblabel "Magnitude (Richter scale)“
set cbrange [0:7]

# Realistic color palette: seismic-style
set palette defined (0  "#0000ff",  2  "#00ffff", 3.5 "#00ff00", 5  "#ffff00",  6.5 "#ff8000", 7  "#ff0000" )
set xrange [28:52]
set yrange [8:32]

# Enable 2D heatmap view (flattened 3D surface)
set pm3d map                    #pm3d stands for "palette-mapped 3D allows you to display a 3D surface or heatma
set pm3d interpolate 10,10      # smooth color transitions
unset key
set border lw 1.5

splot 'quake_data.dat' using 1:2:3 with pm3d
unset output
unset terminal
```

- **Explanation:**
  - title, xlabel, ylabel — describe the plot axes.
  - cblabel — labels the color scale (colorbar).
  - cbrange — limits the colorbar to earthquake magnitudes from 0 to 7.
  - Color Palette
    - Defines a custom color palette mimicking seismic intensity maps:
      - Blue → Cyan → Green → Yellow → Orange → Red
      - Represents low to high intensity visually
      - You can tune the values to match thresholds (e.g., 3 = weak, 5 = moderate, 7 = severe)
  - set pm3d map — enables 2D color mapping (heatmap style).
  - interpolate 10,10 — makes the plot smoother by interpolating 10x10 between data points.
  - unset key — removes the legend, which is unnecessary here.

❑ 3D Plot

```
#x y sin(x)*cos(y)
0 0 0
1 0 0.841471
2 0 0.909297
3 0 0.141120
4 0 -0.756802

0 1 0
1 1 0.454649
2 1 0.490869
3 1 0.076530
4 1 -0.410777

0 2 0
1 2 -0.378401
2 2 -0.408923
3 2 -0.063728
4 2 0.341261

0 3 0
1 3 -0.564642
2 3 -0.610823
3 3 -0.095700
4 3 0.512016

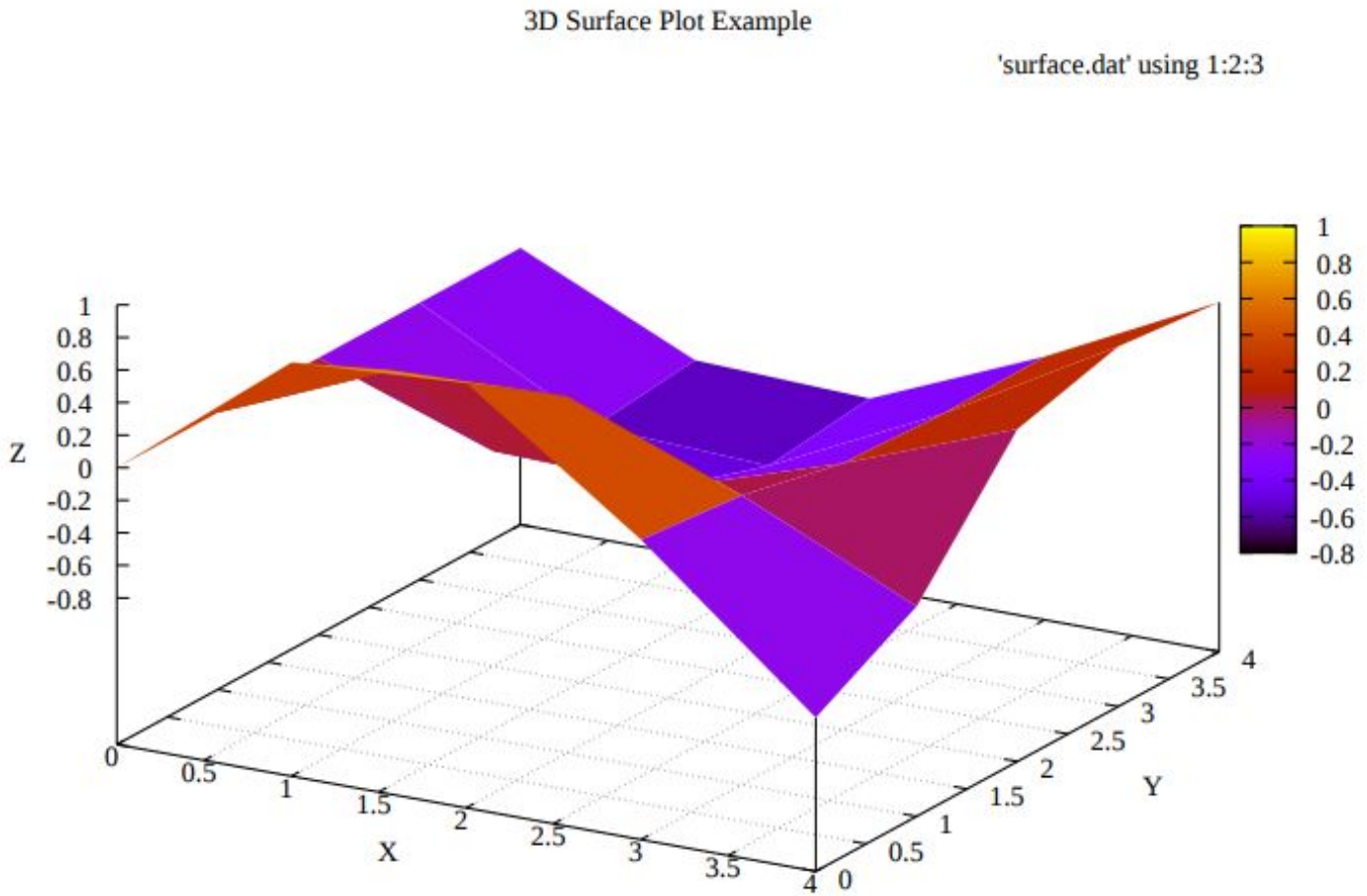
0 4 0
1 4 -0.494864
2 4 -0.535254
3 4 -0.083089
4 4 0.445519
```

- **Sample Code of 3D plot:**

```
set terminal pdfcairo size 20cm, 14cm enhanced font "Times New Roman,16“
set output "surface.pdf“
set title "3D Surface Plot Example“
set xlabel "X”
set ylabel "Y”
set zlabel "Z”
set pm3d
set hidden3d
 set view 60,30  //determines from which angle you “look at” a 3D plot.
 set grid
 splot 'surface.dat' using 1:2:3 with pm3d
 unset output
 unset terminal
```

- **Explanation of code:**
  - set pm3d : Enables palette-mapped 3D surface plotting mode. This mode colors the surface based on the Z values, creating a smooth gradient-colored 3D surface. It requires the input data to be arranged on a grid (points ordered row-wise with blank lines between rows).
  - set hidden3d : Enables hidden surface removal for 3D plots. This makes the plot visually realistic by hiding parts of the surface that are behind other parts when viewed from the current angle.
  - set grid : Adds grid lines to the plot for better visual reference. The grid usually appears on the XY plane but can also help understand the 3D structure.
  - splot 'surface.dat' using 1:2:3 with pm3d : This is the command to create a 3D surface plot. splot is like plot but for 3D data.'surface.dat' is the input data file containing points.
    - using 1:2:3 tells gnuplot to use column 1 as X, column 2 as Y, and column 3 as Z coordinates.
    - with pm3d draws the surface using the palette-mapped colors enabled by set pm3d.

Output:


3D Surface Plot Example
'surface.dat' using 1:2:3
Z
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
X
0
0.5
1
1.5
2
2.5
3
3.5
4
Y
0
0.5
1
1.5
2
2.5
3
3.5
4



Next Topic :Matplotlib